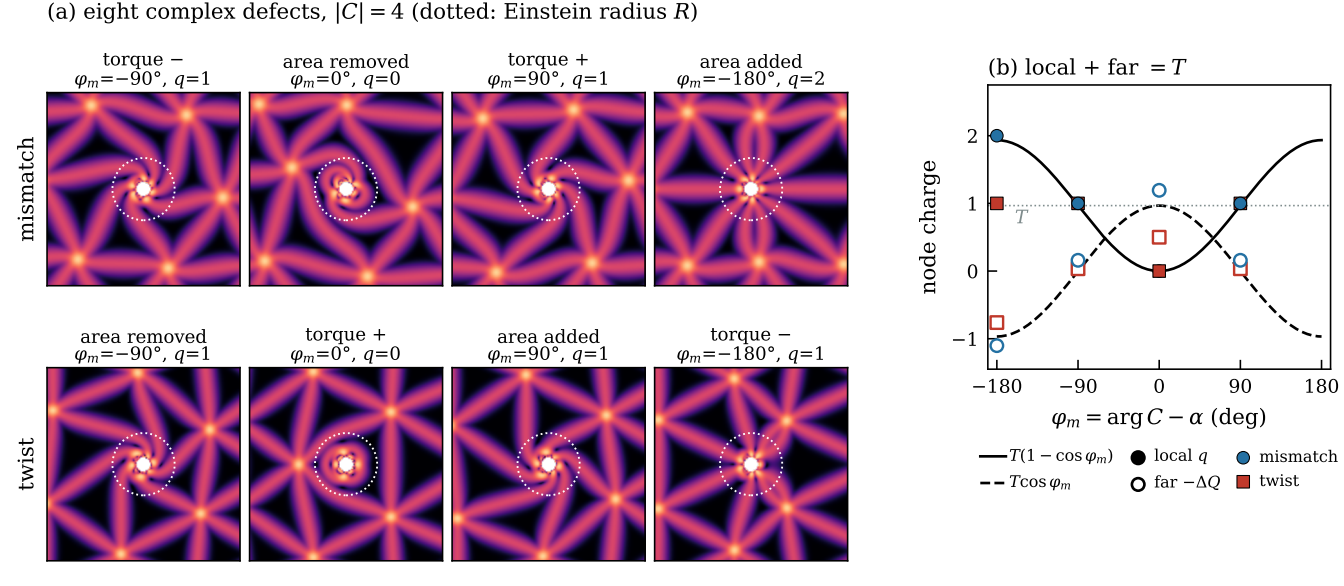

(a) eight complex defects, $|C| = 4$ (dotted: Einstein radius R)
torque −
φ_m=−90°, q=1
area removed
φ_m=0°, q=0
torque +
φ_m=90°, q=1
area added
φ_m=−180°, q=2
mismatch
area removed
φ_m=−90°, q=1
torque +
φ_m=0°, q=0
area added
φ_m=90°, q=1
torque −
φ_m=−180°, q=1
twist
(b) local + far = T
node charge
2
1
0
−1
T
−180
−90
0
90
180
$\varphi_m = \arg C - \alpha$ (deg)
$T(1 - \cos\varphi_m)$
$T\cos\varphi_m$
local q
far $-\Delta Q$
mismatch
twist